

Chapter 15

AI in Human Trafficking Prevention and Safety



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1 Introduction

Human trafficking (HT) remains one of the most pervasive and complex human rights violations in the modern world, affecting millions globally through sexual exploitation, forced labor, child trafficking, and other forms of coercion (United Nations, 2023). Traditional methods of detection and prevention—reliant on manual investigation, victim testimony, and fragmented reporting—have proven insufficient against the evolving tactics of traffickers, who increasingly leverage digital platforms, encrypted communications, and international mobility (United Nations, n.d.). In response, artificial intelligence (AI) has emerged as a robust set of tools that can augment human capabilities to identify trafficking networks, support victims, and disrupt the systems that enable exploitation (Akinsuli, 2022). This chapter is based on a review of 46 peer-reviewed articles published between 2020 and 2025, identified through systematic searches across five major academic databases: EBSCO, PubMed, ProQuest, Scopus, and Web of Science. These sources provided a multidisciplinary foundation for analyzing current trends, technologies, and ethical challenges in the use of AI for anti-trafficking purposes. Table 15.1 summarizes the geographic focus, AI techniques, application domains, and key outcomes of the peer-reviewed studies included in this review.

This chapter explores how AI is being used to address human trafficking across multiple domains. It examines AI-powered techniques for identifying trafficking networks, enhancing law enforcement capacity, and supporting victim recovery. It further interrogates the ethical risks associated with these technologies, including

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Table 15.1 Summary of included studies on AI in human trafficking prevention and safety

Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Aarthee and Ezhilmaran (2020)	India	Agglomerative hierarchical clustering for location, multivariate linear regression for age, OneR algorithm for health symptoms	Analyze risk regions for women crimes	Identified key locations and age groups and map health symptoms affecting sex trafficking victims
Akinsuli (2022)	Global (case studies from USA, UK, international)	Machine learning (ML), natural language processing (NLP), data mining, predictive analytics, facial recognition, data anonymization (data masking, pseudonymization, differential privacy, and federated learning)	Identify and track human trafficking networks; secure victim identities; disrupt illicit activities in real-time	Increased detection rates (Polaris +20%, International Organization for Migration +30%, National Crime Agency +40%), up to 97% anonymization accuracy, disrupted up to 200 networks, reduced response time to 3 h, user satisfaction up to 4.8/5
Anda et al. (2020)	Ireland	ResNet50 (DeepUAge); DCA (deep cropping with artistic proportion) technique	Assist law enforcement in investigating child sexual exploitation material (CSEM), a domain that is often intertwined with child trafficking and exploitation	The model achieves state-of-the-art accuracy (MAE = 2.73 years) for subjects aged 1 to 18
Bermeo et al. (2023)	Ecuador	Review: Supervised, unsupervised, Semisupervised ML	Detect trafficking on social media platforms	Highlights strengths and weaknesses across ML techniques for detection
Bhavanasi and Stylianou (2023)	USA	Object-centric ensemble deep learning	Hotel recognition in trafficking investigations	Outperforms image-only methods on occluded data from hotel-ID dataset

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Table 15.1 (continued)

Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Borrelli and Caltagirone (2020)	USA	ML classifier (Artemis) + kill chain methodology	Disrupt trafficking networks using defense strategies	Artemis classifier offers predictive disruption applicable to other threats
Bouma et al. (2021)	Europe	AI-based anomaly detection + travel pattern recognition	Detect anomalies in travel documents for fraud/trafficking risk	Can extract useful travel behavior patterns; supports faster risk assessment
Deb et al. (2020)	USA	Feature aging module+ feature-to-image aging guidance + face matcher: CosFace	Match children's faces over time: Cross-age face recognition	Improved identification in long-time gaps
Dhinakaran et al. (2021)	India	CNN + crowdsource Mobile app	Real-time child trafficking detection via crowdsourced images	CNN classifies gender, age, impairments; public acts as real-time reporters
Diaz and Panangadan (2020)	USA	NLP + bag-of-words + random Forest (RF)	Identify illegal business ads related to trafficking (yelp reviews)	Random Forest preform best among four models, F1-score 0.77; feature reduction possible; effective for detecting suspicious businesses
Gagneja (2024)	India	ML + big data analytics	Detect patterns in child trafficking/abduction	Identified patterns involving time, location, transportation
Graham (2023) (dissertation)	USA	Comparative analysis of AI platforms (e.g., spotlight, DIG)	Assess law enforcement tools for HT ad detection	AI platforms enhanced efficiency, accuracy, and iterative system design
Granizo et al. (2020)	Ecuador	Text: SVM/naïve Bayes; image: SVM + CNN + Haar features	Detect twitter content promoting child exploitation	Lower quality image and text: Torso-based SVM model outperformed CNN; CNN excelling on clearer images

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Table 15.1 (continued)

Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Hartawan et al. (2023)	Indonesia	Naïve Bayes, random Forest, decision tree	Detect HTSE-related ads via linguistic patterns	RF achieved highest F1 (0.962); detailed algorithm performance comparison
Helderop et al. (2023)	USA (Chicago)	ML classifiers + criminogenic features; logistic regression model vs. artificial neural network (ANN) model	Predict areas of prostitution to identify sex trafficking risks	Spatially lagged logit regression model (ML) had an 81% efficiency; best ANN model (shallow ANN) predicted with 94% accuracy; identifies key correlates of prostitution
Hemadri et al. (2022)	India	CENSER system: MTCNN, MSRHN (ScatterNet + ResNet), cosine similarity, dropout-based uncertainty, false positive filtering	Match trafficked children via images/ sketches	93.17% accuracy on kinship recognition; 12–18fps real-time speed; aided rescue of 13 trafficked children
Hernández- Álvarez et al. (2021)	Not specified	NLP, image processing, Haar filters, SVM	Detect trafficking messages on twitter, identify age/gender	Multiphase system identifies content and demographics using facial geometry
Ing et al. (2023)	France	Decision tree, RF, gradient boosting, SVM, KNN	Detect human Trafficking networks (HTNs) in legal data	Reformulated legal data for better classification; interpretable models provided
Kjellgren (2025)	UK	Web scraping, social network analysis, PCA (Principal component analysis)	Analyze complexity and geographical patterning of online sex networks	Digital traces reveal structure and heterogeneity of online sex networks
Li et al. (2022)	Canada	Rule-based + dictionary + contextual language model	Extract person names from escort ads (NEAT)	NEAT improved F1 scores by 19% over prior methods in name extraction

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Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Li et al. (2023)	USA	Ensemble models: Lexicon models + BERT (embedding model) + Doc2Vec (embedding model)	Detect human trafficking in yelp reviews	Ensemble models(E2/E3) achieved best performance among seven models on accuracy precision and F1 score. Embedding model (M3: BERT) is good at recall
Liu et al. (2023)	Canada	Weak supervision + LLMs + rule-based matching	Extract names from escort ads (SWEET)	Outperformed prior SOTA by 9% F1; released HTGEN dataset
Makka et al. (2025)	India	AI Surveillance system + computer vision	Real-time detection of trafficking and emergencies	Accelerates response times via AI-assisted alerts to police/ hospitals
McDonald et al. (2021)	USA (global fishing industry)	Machine Learning + remote sensing: PU learning, random Forest, bagging, feature engineering	Identify forced labor on fishing vessels	Detected 14–26% of vessels as high-risk; mapped fishing routes and port visits
Moore (2024)	USA	Thematic analysis + data analysis and ML	Analyze traffickers' use of social media algorithms	Suggests algorithms can be exploited; supports proactive intervention strategies
Mukherjee et al. (2021)	USA	ML pipeline (image classifiers): VGG16 + FC for uniform detection; custom CNN for multilabel clothing attribute classification; mask-RCNN for person segmentation	Detect school uniforms for child ID	Supports victim location inference: 96% accuracy for uniform detection; attribute model outperforms random and baseline models on all clothing types

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Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
O’Keefe et al. (2022)	USA	AI-based risk Assessment systems	Detect money laundering linked to HT	Case study shows AI’S power and ethical risks in finance–trafficking link
Ouyang (2023)	USA	NLP (bag-of-words, Word2Vec), ML + agent-based modeling	Track spatiotemporal trends in illicit massage industry (IMI)	Released synthetic datasets; explored language/labor pressures
Perez et al. (2023)	USA	Longformer, XLNet, RoBERTa, ALBERT, BERT, BigBird, GPT-2, CANINE; tokenization: Byte-level BPE, WordPiece, SentencePiece, character-based	Detect HT signals in online sex ads (CSAs): To extract key entities from obfuscated sex ads using transformer-- based NER models for trafficking detection.	Longformer best (F1: 0.815); effective NER in obfuscated CSAs for trafficking detection
Ray et al. (2024)	India–Nepal border, global seafood industry	ML models for cross-context detection	Evaluate HT model applicability in tech-facilitated and human-- facilitated detection	Models adapt across sectors; detection errors remain concern
Razi et al. (2021)	USA	Literature review + data annotation	Classify types of sexual risk (e.g., grooming, trafficking)	93% focus on predator detection; gap in victim prevention approaches
Román and Cuenca (2023)	Ecuador	ML and DL (especially transformer-based)	Detect illicit twitter messages related to trafficking	Found DL methods more adaptive and effective than traditional ML
Saxena et al. (2023)	USA	DeCLUTR-small + style representation	Identify HT vendors in escort ads (IDTraffickers dataset)	Achieved macro-F1 0.8656 (closed-set); r-precision 0.8852 (open-set)

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Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Sharma et al. (2024)	India	Gesture detection + ML	Detect trafficking via behavior analysis in an app	Promotes ML-based gesture analysis as a prevention tool
Summers et al. (2023)	USA	Multi-input deep learning (DL) binary classification models: Recurrent neural networks (RNN) + CNN (EfficientNetB6) + NN; RNN for text classification, convolutional neural networks (CNN, specifically EfficientNetB6 or ENET) for image/emoji classification, and neural networks (NN) for feature classification.	Classify escort ads for sex trafficking (ST) association	Multi-input DL model 5 (ENET + RNN) performs the best among seven models: Was able to correctly identify all 14 of the observations associated with the estimated probability of 0.845 or greater (1.00 precision)
Sunde and Sunde (2021)	Norway	PrevBOT (authorship analysis + ML): Authorship analysis (author profiling and attribution), Stylometry, ML classification on chat features	Detect child sexual exploitation and abuse (CSEA) offenders online	Predictive support for police: ML can identify sexualized speech, false age/ gender, and recidivist CSEA offenders in chatrooms. PrevBOT shows potential as a crime-preventive tool for online CSEA
Tambe and Tambay (2020)	India	AI + Blockchain + digital identity: CNNs (VGG-16, VGG-19, AlexNet, inception), RNN-based activity recognition, object and motion detection in video (video analytics)	Monitor hidden modern slavery: Meet the requirements for supply chain compliance and supervision in Indian enterprises	Enables low-cost continuous monitoring
Thinyane and Sassetti (2020)	Thailand	Expert system (apprise)	Screen for forced labor victims via multilingual AI system	Highlights ethical concerns; stresses need to address pre-existing inequalities

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Table 15.1 (continued)

Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Tobey (2023)	USA	Lexicon-based NLP, logistic regression, random Forest, active learning with deep reinforcement learning	Identify human trafficking risk in yelp reviews of massage businesses; assess business-level risk of IMBs	Developed interpretable models to detect illicit reviews and risky businesses; reinforced learning improved data labeling efficiency; models support scalable, automated law enforcement interventions
Van der Watt (2023)	USA	AI, NLP, Chatbots, ML, data aggregation	To examine how augmented intelligence—Combining human insight with AI, NLP, and chatbots—Can be used to reduce demand for sex trafficking by proactively deterring potential buyers	Chatbots and NLP-based systems deterred sex buyers in 78 U.S. cities; 6000+ predators intercepted; demonstrates scalable, collaborative AI-human counter--trafficking tools
Volodko et al. (2020)	Lithuania	Statistical analysis (initial)	Analyze trafficking indicators in online job ads	Identifies weak but significant correlations; suggests potential for automation
Weinberg et al. (2020)	UK	NLP + AI: Weak supervision for generate labeled training data + supervised learning to classify modern slavery reports + text feature extraction (TF-IDF) to convert report text for machine learning models + N-gram modeling	Analyze modern slavery reports: To automate the assessment of corporate modern slavery statements for legal compliance and supply chain accountability using AI techniques	Improves monitoring compliance

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Table 15.1 (continued)

Study/ authors	Region	AI method(s) used	Application/ goal	Key outcome(s)
Wiriyakun and Kurutach (2021)	Thailand	LIME-based feature selection, naive Bayes (NB), decision tree (DT), neural network (NN; 3-layer, 128 nodes each) on trafficking-10 k (dataset)	Identifying human trafficking-- related advertisements on the internet (especially those for sex trade)	All models' accuracy improved: Accuracy improved to 73% (NB), 72% (DT), and 77% (NN) using LIME
Woolf (2025)	Not specified	General AI tools	Use AI in criminal investigations, risk assessments, victim support	Stresses holistic potential of AI for law enforcement and services
Yagci Sokat et al. (2024)	USA	Predictive Analytics + multitarget metric	Evaluate transit-based anti-trafficking interventions	Early example of predictive evaluation incorporating cost, goals, and org factors
Yao et al. (2021)	China	Random Forest (RF) models using spatiotemporal data	Predict transfer routes and cities for missing persons related to trafficking	High accuracy in predicting destination cities; integrated RF model proves effective
Zhou (2022)	China	Improved SVM with Laplacian regularization, SVM, RBF (Radial Basis Function), KNN (K-Nearest Neighbors); NLP-based feature extraction including keyword patterns and complexity theory	Classify online advertisements for trafficking detection	Effective for law enforcement surveillance: The improved SVM achieved 92% accuracy and AUC (Area Under the Curve), outperforming standard SVM, RBF, and KNN models in identifying trafficking-related online ads

misidentification, data misuse, and the erosion of privacy and dignity. While AI offers unprecedented opportunities for scalability and efficiency, its deployment must be grounded in survivor-centered values, legal accountability, and cross-sector cooperation. By synthesizing findings from recent studies, this chapter aims to provide a critical yet constructive overview of how AI can both illuminate and complicate the global fight against human trafficking.

2 Major AI-Powered Methods for Detecting and Disrupting Human Trafficking

Section 2 outlines the principal categories of artificial intelligence methods used to detect and disrupt human trafficking networks. It is organized into three main parts: Section 2.1 examines machine learning techniques for detecting trafficking patterns across modalities and platforms; Section 2.2 focuses on deep learning and neural network architectures, emphasizing their role in automating visual, textual, and real-time surveillance tasks; Section 2.3 highlights hybrid and ensemble models that integrate multiple approaches to enhance robustness and interpretability; and Section 2.4 synthesizes the geographic and sector-specific applications of these technologies, as well as their observed real-world outcomes.

2.1 *Machine Learning and Trafficking Pattern Detection*

This section focuses specifically on machine learning applications, including multimodal victim identification, spatial prediction of trafficking patterns, and online surveillance of illicit content. These approaches highlight AI's growing role in supporting law enforcement, public health, and corporate accountability efforts worldwide.

2.1.1 Multimodal Detection and Age Estimation Technologies

Recent innovations leverage multimodal machine learning to identify victims of human trafficking in real-world digital environments. Granizo et al. (2020) proposed a combined Support Vector Machine (SVM) and Convolutional Neural Network (CNN) framework to classify tweets and detect minors in images. Specifically, the system classifies tweets based on whether they contain content potentially related to child trafficking—such as suspicious language, hashtags, or user behavior patterns—while concurrently analyzing accompanying images for age estimation. Even when faces are obscured, the model uses upper-body Haar features—a set of mathematically defined visual patterns used to detect structured elements like shoulders and torso contours—to substitute for facial cues in estimating age and gender. Similarly, Deb et al. (2020) introduced a feature aging module that enhances deep face embeddings to improve cross-age identification, crucial for recognizing children years after abduction. Anda et al. (2020) extended age estimation using DeepUAge, a deep learning model based on ResNet50, a 50-layer residual neural network architecture commonly used in image classification tasks. DeepUAge was trained on the VisAGe dataset, a publicly available facial image dataset containing labeled age information, and achieved state-of-the-art results in

underage facial analysis. Together, these studies highlight the potential of Artificial Intelligence (AI) to support law enforcement through robust identification systems that operate effectively across age gaps, varying image quality, and privacy constraints.

2.1.2 Spatial Prediction and Public Health Surveillance

Machine learning has also enabled spatially aware, demographically informed predictions of trafficking patterns. Yao et al. (2021) developed a multimodel random forest system for forecasting likely destination cities of trafficked children in China, combining spatial grids and vector directions to enhance predictive accuracy. In the Indian context, Aarthee and Ezhilmaran (2020) applied hierarchical clustering and regression analysis to identify high-incidence regions and the most vulnerable age groups (women aged 18–30), linking trafficking exposure to health outcomes like PTSD and HIV. In the USA, Helderop et al. (2023) used artificial neural networks with fine-grained urban geocoding to map prostitution activity in Chicago with 94% accuracy, surpassing the accuracy of traditional regression models. These studies demonstrate the utility of machine learning for policy-relevant spatial surveillance and proactive deployment of anti-trafficking resources.

2.1.3 Online Trafficking Detection and Corporate Accountability

Machine learning is increasingly applied to online textual and visual data to detect illicit trafficking-related content and to enhance corporate accountability, that is, ensuring that companies meet their legal and ethical obligations to prevent modern slavery and exploitation within their platforms or supply chains. Summers et al. (2023) built multi-input deep learning models combining EfficientNet—a convolutional neural network optimized for performance and computational efficiency—with bidirectional Recurrent Neural Networks (RNNs) to analyze escort advertisements. The model correctly identified all 14 observations with an estimated probability of 0.845 or higher (precision of 1.00), which is essential for law enforcement, as a lack of precision can lead to wasted effort. This enables prioritizing high-risk ads in resource-constrained enforcement settings. Li et al. (2023) developed a hybrid lexicon-based and embedding-based classification system to detect illicit massage business (IMB) reviews on public platforms, using ensemble methods to balance precision and recall. Diaz and Panangadan (2020) proposed a scalable Random Forest (RF) classifier trained on publicly available reviews from Yelp and Rubmaps—a review platform focused on adult-oriented massage businesses. This model achieved an F1-score of 0.77, indicating a balanced trade-off between precision and recall and reflecting robust classification performance in a domain with highly imbalanced, noisy data. Beyond direct detection, Weinberg et al. (2020) and Tambe (2020) introduced AI frameworks for monitoring modern slavery

compliance in corporate reports and supply chains, respectively, integrating blockchain and human-in-the-loop validation to scale ethical due diligence. These approaches reflect the growing potential of AI for systemic intervention across both public platforms and private sectors.

2.2 *Deep Learning and Neural Networks in Trafficking Detection*

This section delves into the application of deep learning architectures—including CNNs, RNNs, and transformer models—to detect and disrupt human trafficking. Subsections cover visual recognition using CNNs, textual analysis through recurrent and transformer-based models, real-time surveillance systems, and ongoing challenges regarding interpretability and ethics. Together, these approaches demonstrate the power and limitations of deep neural networks in high-stakes and complex trafficking contexts.

2.2.1 Convolutional Neural Networks (CNNs) for Image-Based Identification

Convolutional Neural Networks (CNNs) have emerged as powerful tools in visual-based detection of trafficking victims. These models are especially effective in processing and classifying images, which is crucial for identifying trafficked children or missing persons whose photographs may be outdated or obscured. Anda et al. (2020) introduced DeepUAge, a ResNet50-based model trained on the VisAGE dataset, specifically designed for underage facial images. Compared to existing commercial and academic age estimators—such as Microsoft Azure Face API (mean absolute error [MAE]: 3.60), Amazon Rekognition (MAE: 3.74), and IMDB--WIKI WideResNet (MAE: 20.43)—DeepUAge achieved a significantly lower mean absolute error (MAE) of 2.73, representing a 24.2% improvement over Microsoft Azure and 27% over Amazon Rekognition. This accuracy enables law enforcement to more reliably estimate a subject's age and thus better determine whether an image qualifies as child sexual exploitation material (CSEM), which in turn improves the classification and triage of digital evidence compared to earlier, less specialized models. Similarly, Deb et al. (2020) utilized CNN-based models to implement feature aging techniques, improving cross-age face recognition. Hemadri et al. (2022) introduced a hybrid network that incorporated CNNs to compare present-day photos with sketches and images from earlier years, supporting re-identification over time.

CNNs have also been deployed in mobile-based crowdsourcing applications, such as the system developed by Dhinakaran et al. (2021), which classifies geo-tagged images uploaded by citizens to support the detection of trafficked or exploited

minors. Their model, DesecrateNet, is a 54-layer convolutional neural network designed to identify children under 18 and assess facial impairments or physical disabilities. In performance evaluations, the system achieved an accurate improvement from 92.4% to 96.5%, outperforming popular models such as Inception V3 and MobileNet, while requiring fewer training samples due to efficient use of max pooling and data augmentation techniques. This improvement enables law enforcement to more accurately triage digital evidence, identify minors involved in child sexual exploitation material (CSEM), and prioritize high-risk cases more effectively than with traditional manual screening or generic CNN models. These efforts significantly reduce manual labor for law enforcement and increase the scope of victim identification, particularly in densely populated or resource-constrained regions. Despite their accuracy, CNN-based systems must navigate ethical issues surrounding image storage, consent, and potential misuse. Nonetheless, CNNs remain a cornerstone technology in the domain of trafficking detection, combining automation with high-precision visual analysis.

2.2.2 Recurrent and Transformer Models for Textual Pattern Recognition

While CNNs are optimized for extracting spatial features from images, RNNs are designed to model temporal or sequential dependencies in text or time-series data—making them complementary tools in AI-powered human trafficking detection systems. In the realm of textual analysis, recurrent models such as Recurrent Neural Networks (RNNs)—which process sequential data by retaining information from previous time steps—have enabled significant advances in detecting trafficking--related language online. More recently, transformer-based models (e.g., BERT or Longformer), which rely on self-attention mechanisms rather than recurrence to model contextual relationships, have further improved performance on long and obfuscated text commonly found in illicit online content.

Summers et al. (2023) applied a hybrid deep learning model combining RNNs and CNNs to escort advertisements, capturing temporal and contextual patterns in both textual and visual content. Transformer architectures like BERT and Longformer, as used by Perez et al. (2023), have been particularly effective at handling long, obfuscated sequences common in commercial sex advertisements. These models are trained on large corpora—collections of textual data compiled for natural language processing (NLP) tasks—and fine-tuned for domain-specific tasks such as name extraction, location recognition, and traffic probability classification. Liu et al. (2023) introduced Supervise Weakly for Entity Extraction to fight Trafficking (SWEET), a weak supervision pipeline—an approach that leverages limited labeled data, heuristic rules, and indirect signals rather than fully annotated datasets—for effectively extracting person names from noisy escort advertisements. Similarly, Li et al. (2022) developed Name Extraction Against Trafficking (NEAT), a hybrid Named Entity Recognition (NER) system that merges rule-based and contextual language models to identify ambiguous or

obfuscated names in escort ads, achieving a 19% improvement over previous benchmarks. Their attention mechanisms allow them to focus on semantically rich portions of the text, improving detection accuracy. However, such deep learning applications depend heavily on annotated datasets, often derived from ethically complex sources. Studies conducted in Ecuador (Granizo et al., 2020; Hernández-Álvarez et al., 2021) further demonstrated the utility of transformer models and natural language processing (NLP) pipelines for analyzing Twitter content related to human trafficking, particularly involving minors. These systems employed real-time tweet mining, acronym normalization, and age-group detection using both textual and visual inputs. Bermeo et al. (2023) and Román and Cuenca (2023) contributed insights into Spanish-language trafficking detection systems, analyzing how regional linguistic variation across Latin America affects the accuracy and fairness of transformer-based NLP models. Their work highlights the need for culturally adaptive AI systems in multilingual contexts. Concerns remain about algorithmic bias and transparency, particularly when these models are deployed in law enforcement settings. For example, predictive models may underperform on linguistic or demographic subgroups or produce black-box decisions that are difficult to audit in legal proceedings. These concerns raise essential ethical questions, which are addressed in more detail later in the chapter. Nevertheless, their strong performance on text classification tasks makes them vital in decoding the complex linguistic markers of human trafficking.

2.2.3 Deep Learning in Real-Time Surveillance and Detection Systems

Deep learning models are increasingly integrated into real-time surveillance and emergency response systems to detect trafficking and related crimes. Makka et al. (2025) developed a CNN-based surveillance system capable of identifying incidents such as human trafficking, accidents, and medical emergencies, and automatically sending alerts to nearby authorities. These systems integrate multiple inputs, including video feeds and sensor data, to continuously monitor high-risk areas. Sharma et al. (2024) suggested a data science-based approach that analyzes images — including potential gesture detection — to identify possible human trafficking cases. This approach emphasizes portable, user-facing AI tools that can be deployed in field investigations or emergency reporting apps. In public transit contexts, similar AI models are used to intercept trafficking attempts, as shown in Yagci Sokat et al. (2024). Moreover, projects such as the school uniform recognition pipeline (Mukherjee et al., 2021) use CNNs to help identify trafficking victims by matching clothing patterns in images. Similarly, Hartawan et al. (2023) applied a deep learning ensemble to analyze online sex trade advertisements in Indonesia, demonstrating how linguistic and structural patterns can be leveraged for real-time risk classification and disruption of trafficking-related postings. In another example, Granizo et al. (2020) developed a two-phase real-time detection system that intercepts trafficking-related content on Twitter, identifies suspicious posts, and

estimates user characteristics such as age and gender through image processing and Support Vector Machine (SVM) classification. While highly effective, these models raise significant ethical questions regarding surveillance, data retention, and individual privacy. For example, scraping user images without consent and profiling minors based on their appearance may infringe on individual privacy and data protection norms. As real-time systems gain prominence, developing transparent oversight mechanisms becomes crucial. These concerns are discussed in greater depth later in this chapter. Nonetheless, such applications underscore the transformative potential of deep learning in both proactive and reactive anti-trafficking interventions.

2.2.4 Challenges of Interpretability and Ethical Considerations in Deep Learning

Despite their impressive performance, deep learning models often face criticism for being “black boxes,” making their internal logic difficult to interpret and audit (Hassija et al., 2024). This limitation poses challenges for law enforcement and judicial processes that demand transparency and accountability. Razi et al. (2021), in a systematic review of online sexual risk detection systems, found that many models prioritized algorithmic performance over user-centered evaluation, raising concerns about the societal impacts of such deployments. To address this issue, researchers such as Wiriyakun and Kurutach (2021) advocate integrating explainable artificial intelligence (XAI)—a set of techniques and methodologies that make the outputs and internal logic of AI models interpretable to humans. XAI methods attempt to clarify the decision-making process of deep learning models by visualizing feature importance, simplifying model structures, or generating human-readable explanations.

Additionally, ethical concerns abound in the use of deep learning for trafficking detection, including data privacy, informed consent, surveillance overreach, and the potential criminalization of vulnerable populations. Thinyane and Sasseti (2020) highlighted that AI systems, if not carefully designed, may amplify existing structural inequalities and marginalize vulnerable groups, advocating for a human rights--based approach to system design and deployment. For instance, when models are trained on escort advertisement data or social media content, the ethical sourcing and annotation of such data must be scrutinized. Sunde and Sunde (2021), in developing the PrevBOT tool for early detection of online child sexual exploitation, emphasized the legal tensions between proactive surveillance and ensuring suspects’ fair trial rights, calling for new governance frameworks around predictive AI tools. As these technologies continue to evolve, incorporating interdisciplinary oversight, community input, and robust ethical frameworks is essential to ensure their responsible use in combating human trafficking.

2.3 *Hybrid and Ensemble Approaches in Human Trafficking Detection*

This section explores the growing use of hybrid and ensemble strategies to enhance the effectiveness, adaptability, and transparency of AI systems for trafficking detection. It highlights multi-input models that combine visual, textual, and metadata signals; ensemble learning techniques to improve robustness; rule-based and machine-learning integration; and real-world case studies that demonstrate operational deployment. These approaches aim to bridge the gap between precision and generalizability in applied AI.

2.3.1 Multi-Input Models Combining Text, Image, and Metadata

Multi-input models have proven effective in synthesizing diverse data types—textual, visual, and contextual—for trafficking detection. Summers et al. (2023) developed a hybrid architecture integrating CNNs for visual content, RNNs for sequential text data, and structured metadata inputs, enhancing the detection of trafficking signals in escort advertisements. Such architectures enable the simultaneous processing of emojis, keywords, and geographic information, thereby improving contextual understanding and minimizing false positives. Tobey (2023) developed multi-input models that integrate Natural Language Processing (NLP)-based keyword detection, business-level metrics, and spatial data to assess trafficking risks in illicit massage businesses. These models provide interpretable and actionable insights for resource allocation—such as flagging businesses that use trafficking-related euphemisms in online advertisements, identifying clusters of high-risk locations near transportation hubs, or detecting businesses with unusually long operating hours that suggest exploitative practices. Such insights enable law enforcement agencies and social service organizations to strategically prioritize site visits, allocate investigative resources, and design targeted outreach efforts. Complementing this, Ouyang (2023) combined textual analysis, financial records, and image data to detect structural indicators of labor exploitation within the illicit massage industry, showcasing the practical power of multimodal AI pipelines.

Another example is the school uniform identification system (Mukherjee et al., 2021), which extracts clothing texture, color, and logo patterns from images to match schools. These models reflect the real-world complexity of data and offer a more holistic approach to trafficking detection. However, multi-input models are often resource intensive and require well-curated, annotated datasets for effective training. Their implementation in practice calls for interdisciplinary collaboration among technologists, law enforcement, and civil society to ensure usability and fairness.

2.3.2 Ensemble Learning for Robustness in Trafficking Detection

Ensemble learning, which combines the strengths of multiple models, has shown promise in improving the robustness of trafficking detection systems. Techniques such as bagging, boosting, and voting ensembles aggregate predictions from multiple classifiers, reducing overfitting and enhancing generalizability. For example, Bhavanasi and Stylianou (2023) implemented an ensemble model combining predictions from object-centric classifiers to improve hotel recognition accuracy in trafficking investigations. Rather than analyzing entire images, their approach detects and isolates specific objects commonly found in hotel rooms—such as beds, lamps, artwork, floors, curtains, shelves, and seating—using an open-set object detection model (Grounding DINO). For each object type, a separate deep learning model is trained to recognize hotel-specific patterns based solely on that object (e.g., the design of a headboard or the color and shape of a lamp). At inference time, predictions from each visible object in the query image are aggregated to identify the most likely hotel. This ensemble method significantly improves robustness in real-world scenarios where victim images are heavily occluded, and traditional full-image models fail, enabling investigators to geolocate scenes using partial yet distinctive visual cues. Similarly, Zhou (2022) employed ensemble methods—an approach that combines multiple classifiers to improve predictive performance and robustness—to analyze classified web advertisements and online behavioral data (e.g., advertisement language patterns, frequently used words and phrases, victim descriptions, referenced websites, and geographic indicators) to detect patterns indicative of human trafficking. Hartawan et al. (2023) evaluated several ensemble classifiers, including Random Forest and Decision Trees, to detect trafficking-related linguistic cues in Indonesian online advertisements. The Random Forest model achieved a high F1 score of up to 0.962, indicating a strong balance between precision and recall in identifying trafficking-related language. These results were maintained even in adversarial text environments—contexts where traffickers deliberately obscure meaning through misspellings, slang, emojis, and coded language to evade detection. These models perform exceptionally well in noisy environments characterized by informal, inconsistent, and unstructured advertisement texts. Data heterogeneity posed a challenge due to variations in ad formats, writing styles, and linguistic patterns across platforms, unlike standardized corpora, where language is more uniform. Despite their benefits, ensemble models can be computationally demanding and difficult to interpret, which may limit their transparency and scalability in field settings. Nonetheless, their superior performance makes them valuable assets in law enforcement toolkits, especially for cross-platform and high-stakes detection tasks.

2.3.3 Integration of Rule-Based Systems with Machine Learning

Combining rule-based systems with machine learning is another common hybrid strategy that balances precision with adaptability. Rule-based systems, rooted in domain knowledge and expert heuristics, are effective at capturing explicit trafficking indicators—such as specific keywords, phone numbers, or linguistic patterns. Liu et al. (2023) utilized rule-based tagging in combination with large language models in their SWEET pipeline to extract names from escort ads. Similarly, Li et al. (2022) developed NEAT (Name Extraction Against Trafficking), a hybrid NER system that merges dictionary-based approaches with contextual NLP models to detect person names with improved accuracy. Kjellgren (2025) employed a hybrid analytical approach that combined rule-based classification with social network and component analysis to uncover structural relationships within online off-street sex markets, providing new insights into hidden trafficking dynamics—such as the presence of centrally positioned brokers, overlapping advertisement networks, and geographic clustering of illicit services across urban hubs. These hybrid models are particularly useful when labeled data is sparse or adversarial obfuscation is prevalent, as is the case when traffickers use vague language, coded emojis, or alias switching to evade detection. However, such systems require continuous updating as trafficking language and tactics evolve. They also necessitate close collaboration with domain experts to ensure rule relevance—that is, the continued applicability and accuracy of manually crafted classification rules for detecting evolving trafficking cues—and to prevent bias.

2.3.4 Case Studies: From Escort Ad Analysis to School Uniform Recognition

Several case studies exemplify the practical deployment of hybrid and ensemble methods in real-world settings. The Global Emancipation Network's collaboration with UK police (Mukherjee et al., 2021) automated school identification from photos of trafficked children, using two complementary AI models—one to detect school uniforms and another to classify their attributes. In the context of commercial sex advertisements, hybrid models developed by Summers et al. (2023) and Liu et al. (2023) effectively linked disparate data sources to flag high-risk postings and detect coordinated trafficking activity. These examples demonstrate the value of hybrid systems not only in improving model performance but also in translating technical capabilities into operational insights. Volodko et al. (2020) conducted a case study in Lithuania analyzing over 6000 job ads, in which automated methods flagged 98% of posts as containing at least one labor trafficking indicator, demonstrating the importance of hybrid tools in labor-focused detection. Clearly, these applications underscore the need for human-in-the-loop systems to validate AI outputs and ensure victim-centered approaches.

2.4 Geographic Focus and Observed Outcomes

This final section synthesizes the implementation of AI-powered anti-trafficking technologies across geographic regions and economic sectors. It examines regional applications in South/Southeast Asia and Western countries, specialized use cases in high-risk industries such as fisheries and illicit massage businesses, and assessments of real-world impact. By contextualizing these systems, the section highlights how local needs, infrastructure, and policy landscapes shape both the success and limitations of AI interventions in trafficking contexts.

2.4.1 Regional Applications: South and Southeast Asia

South and Southeast Asia represent prominent focus areas in AI-assisted trafficking detection due to high trafficking prevalence and diverse cross-border dynamics. Aarthee and Ezhilmaran (2020) applied hierarchical clustering and regression analysis to investigate patterns of female trafficking in India, identifying high-risk states and correlating victim demographics with post-trafficking health outcomes. Hemadri et al. (2022) and Gagneja (2024) worked on systems in India incorporating facial recognition and GPS tracking to identify trafficked children and enable rapid intervention. At the India–Nepal border, Ray et al. (2024) developed machine learning models for optimal resource allocation in detecting trafficking in transit zones. In Thailand, the Apprise system (Thinyane & Sasseti, 2020) applied multilingual AI tools to screen potential victims of forced labor. Complementing this, Wiriyakun and Kurutach (2021) used explainable AI techniques—models that provide human-interpretable reasoning for their predictions—to enhance the detection of trafficking-related features in Thai online sex advertisements, thereby improving the interpretability of predictive models. In Indonesia, Hartawan et al. (2023) employed an ensemble of machine learning models to classify sex trade advertisements, demonstrating the region’s growing use of AI to detect online forms of trafficking. These regional initiatives emphasize early intervention and cross-border coordination, while also raising essential questions about infrastructure and data accessibility in low-resource settings—challenges that are further explored in the ethical considerations and limitations section.

2.4.2 Western Contexts: USA, Canada, and Europe

In Western contexts, AI systems are often applied to digital surveillance, criminal investigations, and corporate accountability. In the USA, platforms such as TraffickCam and IDTraffickers (Graham, 2023; Saxena et al., 2023) support law enforcement by linking escort ads and hotel images to known trafficking cases. Perez et al. (2023) used Longformer-based NLP models to improve the detection of obfuscated language in commercial sex advertisements, while Razi et al. (2021) reviewed online sexual exploitation detection models, criticizing the lack of

victim-centered evaluations. In Canada, Liu et al. (2023) and Li et al. (2022) developed SWEET and NEAT to extract person names from escort ads using hybrid NER models. In Ireland, Anda et al. (2020) proposed DeepUAge for more accurate age estimation in CSEM investigations. In Lithuania, Volodko et al. (2020) identified labor trafficking patterns in job ads, showing that AI can reveal exploitation indicators in the formal labor market, such as subminimum wages, excessive hours, and high worker dependence through accommodation and wage deductions. Across the UK and continental Europe, technologies such as blockchain-based monitoring systems that trace labor supply chains for ethical sourcing compliance (Tambe & Tambay, 2020) and document authentication tools using semantic text analysis to verify migrant work permits (Bouma et al., 2021) have been proposed to identify trafficking within corporate and cross-border infrastructures. While these efforts reflect a maturing technical landscape, they also raise emerging concerns about civil liberties and ethical AI use—issues that are further discussed later in this chapter, in the section on challenges and limitations.

2.4.3 Sector-Specific Hotspots: Fisheries, Construction, and IMBs

Certain industries have been identified as hotspots for trafficking, prompting the development of specialized AI detection tools. McDonald et al. (2021) used satellite vessel monitoring data and machine learning to identify high-risk fishing vessels engaged in forced labor. In the illicit massage business (IMB) sector, Tobey (2023) and Ouyang (2023) developed hybrid models integrating business metadata, customer reviews, geospatial features, and financial patterns to assess trafficking risks. Financial surveillance offers another lens: O’Keefe et al. (2022) proposed integrating AI into regulatory systems to detect patterns linked to trafficking operations in banking and compliance data. Tambe and Tambay (2020) developed a blockchain-based due diligence tool to identify signs of modern slavery in UK supply chains. These industry-specific applications demonstrate how AI can adapt to the unique features, risks, and data structures of different sectors. However, their real-world effectiveness depends on interorganizational coordination—such as data sharing between regulators, law enforcement, and private entities—and robust policy alignment to ensure that AI-generated alerts translate into legal action or protective measures. Without such integration, AI tools risk becoming siloed or underutilized, limiting their impact on actual trafficking prevention and prosecution.

2.4.4 Impact Assessment: Rescue Rates, Legal Outcomes, and Policy Uptake

While technical progress in AI models is well documented, their real-world effectiveness remains poorly evaluated in terms of rescue outcomes, legal accountability, and policy integration. Hemadri et al. (2022) noted that AI-assisted facial recognition tools helped law enforcement act swiftly on trafficking suspicions, highlighting

their potential operational impact. In Thailand, Thinyane and Sassetti (2020) reported improved multilingual screening of likely trafficking victims using Apprise. Weinberg et al. (2020) analyzed corporate compliance with modern slavery legislation and demonstrated that AI-supported monitoring improved reporting quality. Graham (2023) assessed tools like TraffickCam and Domain Insight Graph (DIG), calling for more rigorous performance validation in law enforcement settings. Finally, Borrelli and Caltagirone (2020) introduced the human trafficking “kill chain” model and advocated for centralizing data collection, promoting cross-sector collaboration, and embedding cybersecurity principles into AI deployment. These examples illustrate both the promise and the current limitations of AI technologies in influencing systemic change and improving outcomes for trafficking survivors.

3 AI in Law Enforcement and Victim Support

3.1 Enhancing Investigative Intelligence and Surveillance Capacity

AI technologies are transforming how law enforcement agencies gather, process, and act on intelligence related to human trafficking. By automating surveillance, synthesizing fragmented digital evidence, and revealing hidden network structures, AI enables more proactive, targeted, and timely interventions. This section discusses key domains where AI augments investigative capabilities.

3.1.1 Network Analysis and Intelligence Augmentation

Several studies apply AI to unravel hidden trafficking networks embedded in online platforms and physical operations, such as off-street sex markets, digital escort services, and loosely connected transit hubs. Kjellgren (2025) used principal component analysis and web scraping to analyze over 15,000 UK-based off-street sex networks, identifying patterns of co-occurrence among actors and services. While not focused on real-time detection, Ing et al. (2023) demonstrate how AI can be used to extract and visualize relational patterns from legal case data, offering retrospective insights into trafficking networks based on confirmed judicial evidence. Ing et al. (2023) employed natural language processing and vector-based classification models to analyze legal cases in France, constructing interpretable knowledge graphs that reveal relational links among suspects, locations, and victim profiles. These approaches help identify organized criminal structures that may otherwise remain obscured through manual casework. Furthermore, Graham (2023) evaluated platforms such as the Domain Insight Graph (DIG), which uses AI to connect disparate digital indicators across escort ads, hotel data, and user behavior, offering law enforcement a visual interface for real-time intelligence tracking. These tools enable

officers to spot evolving trafficking networks, even when their digital footprints are intentionally fragmented across jurisdictions or platforms.

3.1.2 Surveillance, Early Warning, and Predictive Monitoring

AI-enhanced surveillance systems provide continuous monitoring and early warning capabilities that support trafficking intervention. Makka et al. (2025) developed a real-time CNN-based surveillance system that detects human trafficking and emergency incidents through public camera feeds, generating alerts for nearby responders. In the maritime domain, McDonald et al. (2021) leveraged satellite data and machine learning to identify fishing vessels likely engaged in forced labor based on movement patterns and labor ratios. Thinyane and Sasseti (2020) introduced the Apprise tool, a multilingual expert system used at checkpoints in Thailand to screen potential victims using smartphones, enabling frontline responders to access AI-enhanced risk assessments. Sharma et al. (2024) conceptually proposed incorporating gesture detection into an image analysis framework to help identify potential indicators of human trafficking, although this functionality was not yet implemented in their prototype. These systems exemplify how AI can support proactive and reactive interventions by monitoring risky environments and supporting frontline decision-making.

3.1.3 Document Authentication and Systemic Coordination

AI also supports investigative infrastructure by enhancing document verification and enabling multiagency collaboration. Bouma et al. (2021) proposed using multimodal machine learning to verify travel and identity documents, thereby improving the detection of fraudulent or forged records at border crossings. Meanwhile, Borrelli and Caltagirone (2020) proposed a “human trafficking kill chain” model, adapted from military cyber defense, to guide inter-agency collaboration and centralized intelligence sharing. The model emphasizes a prevention-oriented approach that integrates AI-enabled signals across the trafficking timeline—from recruitment to transport to exploitation. Collectively, these technologies expand law enforcement’s operational capacity from reactive casework to predictive, intelligence-led intervention. However, concerns remain around due process, data privacy, surveillance overreach, and the potential for algorithmic bias—especially when AI tools are used in real-time decision-making or inter-agency data sharing. These risks raise critical questions about transparency and accountability in AI-driven policing.

3.2 *Supporting Victim Identification and Recovery*

AI systems currently play a vital role in assisting law enforcement and service providers in identifying, locating, and supporting victims of trafficking. From facial recognition technologies to multilingual screening tools, AI facilitates more efficient interventions while raising critical ethical questions regarding privacy, consent, and justice. Key concerns include the risk of misidentifying victims through biased algorithms, the use of surveillance tools without informed consent, and the potential retraumatization or criminalization of victims during automated data processing. These challenges underscore the importance of ensuring that AI deployment aligns with victim-centered principles and legal safeguards. This section outlines key technological approaches and their practical contributions to victim-centered strategies.

3.2.1 Facial Recognition and Image Matching Technologies

Facial recognition technologies are increasingly used to match current images of missing or trafficked individuals with past photos, sketches, or kinship-based reconstructions. Hemadri et al. (2022) developed the ChildrEN SafEty and Rescue (CENSER) system using the Augmented ScatterNet ResNet Hybrid (MSRHN) framework, enabling comparisons between current-day images and older or partial images, including sketches provided by families.

Deb et al. (2020) advanced cross-age face recognition by introducing a feature aging module that learns to project facial features into deep space, thereby enhancing recognition of children whose appearance has changed over time. Their system incorporates both feature-level transformation and image-level synthesis to simulate facial aging while preserving identity. Tested on child-specific datasets such as Child Face Aging (CFA) and In-the-Wild Child Celebrity (ITWCC), their model boosted CosFace's open-set identification accuracy from 22.91% to 25.04% when the time gap exceeded 10 years. Moreover, their approach outperformed commercial matchers on benchmark datasets such as FG-NET and CACD-VS, achieving rank-1 accuracies of 95.91% on FG-NET and 99.58% on CACD-VS. These results demonstrate significant advancement in long-term re-identification of missing children—an area where conventional models often fail due to aging variability.

Anda et al. (2020) contributed DeepUAge, a ResNet50-based deep learning model trained on a large-scale underage facial dataset (VisAGE) for precise age estimation in child sexual exploitation material (CSEM) investigations. Their system achieved a mean absolute error (MAE) of 2.73 years, outperforming Microsoft Azure and Amazon Rekognition services. Unlike generic cloud-based age predictors, DeepUAge was specifically tailored for underage faces and optimized for

forensic use. The authors also developed a novel offline facial preprocessing technique (DCA) suitable for privacy-sensitive law enforcement contexts, enabling better alignment and accuracy on images collected from diverse, unconstrained sources.

These technologies improve investigative efficiency, especially in large-scale datasets where manual identification would be infeasible. Mukherjee et al. (2021), in collaboration with UK police, implemented an automated school uniform recognition system that uses CNNs to identify educational institutions from images of children, further aiding in victim tracing and contextual identification. Together, these visual-based approaches offer scalable tools for locating victims—especially minors—when other forms of identification are unavailable or less than optimal.

3.2.2 Natural Language Processing and Name Recognition

Text-based detection tools also support victim identification, particularly within the digital sex trade. Liu et al. (2023) introduced SWEET, a weak supervision pipeline—an approach that leverages limited labeled data, heuristic rules, and indirect signals rather than fully annotated datasets—that combines rule-based and language model-based tagging to extract names from escort advertisements. Li et al. (2022) proposed NEAT, a hybrid NER system that merges dictionary methods and contextual embedding models to improve person name extraction, even when identifiers are obfuscated. These tools not only aid law enforcement in linking cases across platforms but also help reunite victims with families by identifying aliases and coded references. Thinyane and Sassetti's (2020) case study of the Apprise platform offers a multilingual, mobile-based expert system for screening potential victims of human trafficking and forced labor in field settings—such as seafood factories, industrial zones, and informal labor sites—where privacy, trust, and communication barriers often hinder victim identification. By integrating AI-driven question flow and language translation, Apprise helps non-specialist frontline workers—such as labor inspectors, nongovernmental organization (NGO) staff, and intergovernmental organization (IGO) personnel—identify victims of forced labor or sexual exploitation, even when they lack formal training in trafficking detection.

Wiriayakun and Kurutach (2021) extended this work by applying explainable AI Local Interpretable Model-Agnostic Explanations (LIME) to the classification of online escort advertisements in Thailand—specifically from the trafficking-10 k dataset. Their explainable artificial Intelligence (XAI)-based feature selection method improved interpretability by identifying and ranking important keywords influencing classification outcomes, while also enhancing model accuracy across Naive Bayes (from 68% to 73%), Decision Tree (71% to 72%), and Neural Network models (72% to 77%) compared to a prior baseline.

3.2.3 Ethical Considerations in Victim-Centered AI Design

Despite their potential, these technologies raise critical ethical concerns. Facial recognition models can misidentify individuals, especially across and within racial or age groups, and their use without consent can retraumatize survivors. Projects like PrevBOT (Sunde & Sunde, 2021), although aimed at identifying online offenders, underscore the legal dilemmas surrounding data use and suspects' rights to a fair trial. These include concerns about privacy violations through the automated processing of personal and biometric data, the risk of police provocation if AI avatars pose as children, and the admissibility of chat logs as evidence without violating the suspect's right against self-incrimination. While PrevBOT operates under human oversight, its deployment must comply with strict safeguards under the EU Law Enforcement Directive and the ECHR to ensure legality and fairness. Razi et al. (2021) emphasized that many existing tools prioritize detection efficiency over user-centered outcomes, suggesting a need for frameworks that evaluate tools from victims' perspectives. To ensure that AI-driven systems genuinely support both recovery and dignity for survivors, and also contribute to the identification and apprehension of perpetrators to prevent further harm, interdisciplinary governance is essential. This includes oversight from legal, social work, and survivor-led organizations, as well as technologists. As AI becomes embedded in frontline work, systems must balance predictive accuracy with fairness, transparency, justice, and the protection of survivors' rights.

3.3 Demand-Side Disruption and Preventive Engagement

While most anti-trafficking interventions focus on identifying victims and prosecuting traffickers, AI is also being employed to address demand-side dynamics—such as identifying exploiters, disrupting illicit markets, and preventing future offenses. These tools offer new opportunities for proactive engagement, deterrence, and systemic reform, but also require careful consideration of civil liberties, transparency, and unintended consequences.

3.3.1 Offender Profiling and Online Predator Detection

AI technologies are increasingly used to monitor online environments for exploitative behavior and to identify individuals at risk of committing trafficking-related offenses. Sunde and Sunde (2021) developed PrevBOT, a proactive AI tool designed to detect potential child sexual exploitation offenders in Norway by analyzing digital behavior patterns, such as linguistic cues, age deception, and attempts to move conversations to private channels. The system incorporates ethical and legal safeguards, including data minimization and fair trial

protections—such as avoiding entrapment, ensuring human oversight in decision-making, and restricting the use of chat logs unless suspects initiate contact—to ensure human rights compliance in predictive law enforcement. Similarly, Razi et al. (2021) reviewed a broad range of online predator detection systems in the USA, critiquing the lack of victim-centered evaluations and calling for more socially responsible model development.

These efforts reflect a shift from reactive investigation to preventative policing, offering tools to intervene before harm occurs. However, they also raise significant concerns regarding false positives, due process, and the stigmatization of individuals based on probabilistic modeling. Ethical considerations include the risk of reinforcing bias in training data, lack of transparency in algorithmic decision-making, and insufficient mechanisms for accountability or redress when errors occur.

3.3.2 Disrupting Illicit Markets and Commercial Platforms

Several AI systems target the platforms and commercial ecosystems that facilitate trafficking. Liu et al. (2023), Li et al. (2022), and Perez et al. (2023) developed tools to monitor and extract information from escort advertisements, helping identify trafficking-related patterns and language. Tobey (2023) and Ouyang (2023) focused on the illicit massage business (IMB) sector, leveraging a combination of geospatial, financial, and lexical indicators to flag suspicious establishments and inform regulatory action. Geospatial features included proximity to truck stops, military bases, and highway exits; financial indicators involved anomalous business patterns such as high cash flow or frequent late operating hours; and lexical features drew on terms in online reviews—such as euphemistic phrases or references to ethnicity—that are statistically associated with commercial sex activity.

Graham (2023) evaluated the Domain Insight Graph (DIG), a user-friendly platform that enables law enforcement agencies to analyze patterns in online sex advertisements—including obscure data such as special characters and phone numbers—using cluster queries to help uncover trafficking operations. Sharma et al. (2024) developed a machine learning-based web application that analyzes hotel room images to detect potential trafficking indicators. Although the authors mention gesture detection as a possible analytical component, their system remains a technical prototype rather than a community-facing or real-time reporting tool.

These models broaden the focus beyond individual actors to include structural and infrastructural components of the trafficking economy, facilitating systemic disruption alongside case-by-case prosecution.

3.3.3 Transparency, Accountability, and Public Participation

Demand-side interventions must be governed by principles of transparency and community accountability. Dhinakaran et al. (2021) proposed a citizen-focused mobile app that allows users to report suspicious activities by submitting geotagged

images, which are processed by AI models for signs of exploitation—for example, indicators such as visible physical impairments, facial scars, or signs of forced child begging. This participatory approach democratizes trafficking detection and enhances early community-based intervention.

Weinberg et al. (2020) developed Project Artificial Intelligence against Modern Slavery (AIMS) to help businesses monitor their supply chains of goods and services—such as seafood from Thailand, electronics from Malaysia and China, textiles from India, and wood from Brazil—for human rights violations, offering dashboards for transparency, risk mapping, and compliance tracking. Tambe and Tambay (2020) proposed integrating AI and blockchain in corporate due diligence processes to address forced labor in supply chains. These initiatives highlight how AI can serve as a tool not just for detection but also for accountability and prevention through structural transformation. For example, their proposed framework combines a self-sovereign digital identity system to verify worker age and consent at hiring, an AI-based monitoring system that analyzes workplace CCTV and payroll data to flag excessive hours, unsafe conditions, or wage violations, and a permissioned blockchain to store verified incidents and evidence in an immutable, transparent ledger accessible to regulators and supply chain partners.

As AI tools become embedded in commercial, legal, and civic infrastructures, it is critical to ensure public transparency, institutional oversight, and survivor-informed design to prevent misuse—such as invasive surveillance of vulnerable populations or biased decision-making that reinforces discrimination—and to enhance long-term impact.

4 Ethical and Privacy Challenges

4.1 *Risks of Misidentification and Bias*

AI systems used in trafficking detection are vulnerable to algorithmic bias, misidentification, and unfair profiling—risks that disproportionately affect marginalized populations (Anda et al., 2020; Deb et al., 2020; Razi et al., 2021; Thinyane & Sassetti, 2020). These risks are particularly acute in face recognition systems and natural language processing tools when datasets are limited, imbalanced, or nonrepresentative.

4.1.1 Facial Recognition Bias and Cross-Age Inaccuracy

Facial recognition tools used to identify victims across age and time are susceptible to errors. Deb et al. (2020) acknowledged that cross-age face recognition can degrade when dealing with older photos, especially in nonwhite populations. Anda et al. (2020) addressed this issue with DeepUAge, a system trained on underage

facial data for CSEM investigations. Still, even this model faces challenges in distinguishing minors across ethnicities and varying image quality. Misidentification not only delays victim recovery but may also lead to wrongful targeting, retraumatization, or privacy violations.

4.1.2 Discriminatory Patterns in Text-Based AI Models

Textual models can also embed bias, particularly when trained on skewed or adversarial datasets—for example, corpora dominated by language from a single demographic or region, which underrepresents other groups (skewed), or datasets intentionally seeded with misleading, discriminatory, or inflammatory content to influence model behavior (adversarial). Razi et al. (2021) observed that many online predator-detection systems emphasize offender profiling while failing to assess the impact on misclassified individuals and innocent users. NLP models used to classify escort ads (e.g., Li et al., 2022; Liu et al., 2023; Perez et al., 2023) may reinforce linguistic and cultural stereotypes—especially when relying on obfuscated or coded language common in marginalized communities.

4.1.3 Need for Explainability and Human Oversight

The use of “black-box” models in high-stakes environments, such as law enforcement, has prompted calls for greater transparency. Wiriyakun and Kurutach (2021) demonstrated that explainable AI (XAI) can improve the interpretability of classification models used in the Thai sex trade context by using Local Interpretable Model-Agnostic Explanations (LIME) to identify and rank the specific words in online escort advertisements that most influence the model’s decision. By showing the contribution of each word to a “trafficking” or “not trafficking” classification, investigators and law enforcement can understand why the model reached a given outcome, making the decision process more transparent, fostering trust in automated detection, and enabling accountability in model deployment. Sunde and Sunde (2021) echoed this concern in the PrevBOT system, advocating for legal safeguards to protect fair trial rights. Across systems, explainability and human-in-the-loop review processes are essential to counteract error propagation and ensure ethical AI implementation.

In sum, the risk of algorithmic bias and misidentification remains a fundamental ethical challenge in applying AI to trafficking detection. Addressing it requires not only diversified training data, explainable models, and robust mechanisms for redress when harms occur, but also well-defined training protocols, independent oversight, and procedural guardrails to ensure consistent, accountable, and fair system performance over time.

4.2 *Data Misuse and Security Concerns*

The deployment of AI systems in anti-trafficking contexts involves handling highly sensitive data, including personal images, behavioral patterns, financial records, and location traces. While such data are vital for detection and intervention, its misuse—either through inadequate security, lack of consent, or overreach—can endanger victims, violate privacy, and erode trust in interventions (Diaz & Panangadan, 2020; Gagneja, 2024; Hemadri et al., 2022; Woolf, 2025; Zhou, 2022).

4.2.1 Sensitive and Ethically Sourced Data

Many AI systems rely on data scraped from escort advertisements, job platforms, social media, or public reviews—often without explicit consent from the individuals involved. Li et al. (2022), Liu et al. (2023), and Perez et al. (2023) use obfuscated language and indirect identifiers from commercial sex ads to train detection models, such as deliberate misspellings (“mizz” for “miss”), symbolic substitutions (e.g., a rose emoji to denote payment), or using names that double as common nouns (“Amber,” “Hazel,” etc.) to disguise individuals’ identities. Although these datasets are rich in signals, they risk misclassifying consensual sex workers or exposing vulnerable individuals to law enforcement surveillance. Razi et al. (2021) and Thinyane and Sasseti (2020) cautioned that using such data without strong ethical frameworks may reproduce structural inequalities and undermine survivor dignity.

4.2.2 Surveillance Overreach and Community Risk

AI-based tools such as Dhinakaran et al.’s (2021) citizen reporting app and Sharma et al.’s (2024) mobile gesture detection model raise concerns about surveillance overreach and misuse. While enabling public participation in trafficking detection, these tools can be misused to report false positives or racially profile individuals in public spaces. Furthermore, McDonald et al.’s (2021) satellite monitoring of fishing vessels, while effective, raises legal questions about the right to commercial privacy in unregulated zones, among others. These concerns underscore the need for robust governance, especially when technologies are deployed outside formal regulatory structures.

4.2.3 Data Security, Retention, and Anonymization

Storing and managing sensitive data—such as facial images of minors (Anda et al., 2020; Hemadri et al., 2022), or financial transactions (O’Keefe et al., 2022)—requires strict security protocols and anonymization standards. Many current systems lack clear guidelines on data retention periods, audit trails, or mechanisms for data subject

redress. Weinberg et al. (2020) and Tambe and Tambay (2020) proposed transparency-enhancing tools in corporate and supply chain contexts, but similar measures are needed across all AI applications handling personal and biometric data.

As AI tools grow more pervasive in the anti-trafficking ecosystem, ensuring responsible data stewardship becomes an ethical imperative. This includes building privacy-by-design architectures—systems that integrate privacy safeguards from the outset, such as anonymizing personal identifiers before storage or processing, and limiting access through role-based permissions—enforcing secure data-handling standards, and empowering individuals and communities with rights over how their data is used.

4.3 Cross-Sector Collaboration and Accountability

Effective and ethical use of AI in combating trafficking requires coordination across sectors—law enforcement, civil society, technology developers, policy makers, and affected communities. Cross-sector collaboration can enhance system design, improve oversight, and ensure that AI tools align with both operational goals and human rights principles. However, such a partnership must also clarify lines of accountability, especially when AI systems produce (or have the potential to create) harmful outcomes or operate across jurisdictions. Van der Watt (2023) examined how augmented intelligence can be integrated into anti-trafficking strategies through cross-sector collaboration involving law enforcement, civil society, technology developers, policymakers, and affected communities. For example, in the Street Grace “Transaction Intercept” initiative, technology developers built an AI chatbot that mimics underage personas to engage suspected buyers online, while law enforcement monitors conversations in real time and can intervene to make arrests; civil society organizations coordinate the outreach component and, informed by survivor input, craft deterrence messages that are both realistic and respectful; and policymakers draw on the operational data to design privacy regulations and accountability frameworks. He underscores that such collaboration must include clear lines of accountability, ethical safeguards, and survivor-informed practices, ensuring that technological innovation advances justice and human rights rather than producing harmful outcomes.

4.3.1 Interagency Intelligence Sharing and Standardization

AI systems function best when supported by a coherent data infrastructure and interagency cooperation. Borrelli and Caltagirone (2020) proposed a “human trafficking kill chain” model that applies cybersecurity principles to map and disrupt trafficking stages, requiring intelligence sharing between local law enforcement, federal agencies, and nongovernmental organizations (NGOs). Graham’s (2023) evaluation of platforms such as TraffickCam and DIG highlighted the benefits of

integrated data systems for improving investigative coordination across state lines. However, differing standards for data access, evidence admissibility, and privacy can hamper collaboration.

Bouma et al. (2021) presented a suite of AI-enabled, multisource techniques for authenticating travel and breeder documents—including visual security feature analysis, breeder document verification, travel pattern extraction, blockchain-based inspection records, and ePassport chip security checks—designed for integration into standardized border control systems. By aligning with internationally recognized specifications, such as the International Civil Aviation Organization (ICAO) ePassport standards, these approaches promote the development of interoperable protocols—that is, common technical rules and data exchange guidelines (e.g., standardized chip data structures, unified scanner output formats, and consistent meta-data fields) that enable different agencies and national systems to interpret and use each other’s verification results. Such technical alignment facilitates cross-border information sharing and operational consistency, supporting interagency collaboration in transnational security contexts.

4.3.2 Corporate Responsibility and Supply Chain Monitoring

As noted in Sect. 2.1.3, AI tools have also been applied in the private sector to strengthen corporate accountability and identify trafficking risks within global supply chains. For example, Weinberg et al. (2020) developed Project AIMS to provide dashboards and risk maps that enable companies to monitor for modern slavery violations, while Tambe and Tambay (2020) proposed integrating AI with blockchain and self-sovereign identity tools to automate corporate due diligence and increase transparency. Moore (2024) examined the institutional and legal constraints of algorithmic audits in corporate anti-trafficking compliance, arguing that current frameworks lack enforceable transparency and accountability provisions, leaving affected workers without adequate protections. Collectively, these tools encourage firms to move beyond compliance checklists toward proactive detection and remediation.

However, voluntary adoption remains uneven—global benchmarks show that over 80% of companies fail to meet even fundamental human rights due diligence indicators, and only 6% have fully implemented all core steps (World Benchmarking Alliance [WBA], n.d.-a). Public accountability mechanisms are also scarce: in 2023, just 27% of companies engaged directly with rights-holders during due diligence processes (WBA, n.d.-b). These low adoption rates, despite years of nonbinding guidelines, illustrate the limits of voluntary action; without regulatory mandates or public oversight, corporate-led AI systems are likely to prioritize reputational risk management over substantive worker protection.

4.3.3 Survivor-Led Oversight and Participatory Design

Centering survivors in AI system design is essential for building ethically grounded tools. Thinyane and Sasseti (2020) emphasized a rights-based approach that foregrounds survivor agency and context-sensitive screening methods. Sunde and Sunde (2022) incorporated legal safeguards and fair trial protections into the PrevBOT system (e.g., ensuring meaningful human oversight in decision-making, strict limits on biometric data processing under the Law Enforcement Directive, safeguards against police provocation, protection against self-incrimination, and clear data retention/deletion rules), demonstrating that accountability mechanisms can be embedded into technical design. Dhinakaran et al.'s (2021) participatory mobile app model further illustrates how community members can contribute meaningfully to anti-trafficking efforts when provided with safe, structured tools, clear usage guidelines, and appropriate training to ensure effective and responsible participation. Woolf (2025) advocated for trauma-informed AI design grounded in survivor-led participation. Her research emphasizes that codesign with trafficking survivors can help mitigate retraumatization risks and improve trust in AI systems designed to support vulnerable communities.

Ensuring accountability in AI deployment requires robust governance frameworks, community engagement, and multistakeholder audits. As AI becomes more embedded in anti-trafficking strategies, aligning technological innovation with justice-oriented values remains an urgent priority.

5 Conclusion

Artificial intelligence is rapidly transforming the global response to human trafficking. From enhancing surveillance and investigative capabilities to identifying victims and monitoring supply chains, AI technologies offer innovative ways to detect, disrupt, and ultimately prevent exploitation. This chapter has shown how AI can illuminate hidden trafficking networks, empower law enforcement, and extend support to victims across diverse sectors and regions. It has also highlighted the critical importance of ensuring that these tools are used responsibly.

Despite the promise of AI, the ethical risks are considerable. Misidentification, algorithmic bias, data misuse, and insufficient accountability can undermine the very protections these technologies seek to provide. As trafficking becomes increasingly digital, so too must our ethical and legal frameworks evolve. Achieving effective, just, and humane anti-trafficking outcomes will require not only technical innovation but also survivor-informed governance, multisector collaboration, and a commitment to transparency, equity, and human rights.

Looking ahead, AI's role in anti-trafficking efforts should be guided by a dual imperative: to enhance the precision and reach of intervention systems and to safeguard the dignity, autonomy, and safety of those most affected. With careful design

and responsible implementation, AI can become a vital ally in the ongoing struggle to dismantle trafficking networks and support those they harm.

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Part III
AI and Social Justice in Changing
Contexts